Chemistry PYQ-Pattern Practice

Original exam-pattern practice questions; not verbatim past-paper copies

Section A — Concept Questions

• Which quantity remains the same for all isotopes of an element? A) mass number B) atomic number C) neutron number D) nucleon number

• Which principle explains the arrangement of electron pairs around a central atom? A) VSEPR B) Boyle C) Hess D) Raoult

• Oxidation is best described as: A) gain of electrons B) loss of electrons C) gain of neutrons D) loss of protons

• Which quantity is required in kelvin for ideal-gas calculations? A) pressure B) volume C) temperature D) mass

• Which bond is formed by sidewise overlap? A) sigma B) pi C) ionic D) coordinate only

Section B — Numerical/Reasoning

• A sample contains 0.50 mol solute in 2.0 L solution. Find its molarity.

• A gas occupies 5 L at one condition. If pressure is doubled at constant temperature, predict the new volume.

• A compound contains elements in a 1:2 simplest atom ratio. Explain how you would determine its empirical formula.

• A solution of known molarity is diluted to twice its original volume. What happens to its molarity?

• An electrochemical cell has a reaction in which electrons are released at one electrode. Identify the process at that electrode.

Section C — Organic/Reaction Reasoning

• Why are alkenes generally more reactive toward addition reactions than alkanes?

• Explain how resonance can stabilize a charged intermediate.

• Differentiate substitution and addition reactions with one representative example each.

• Why can molecular geometry affect polarity even when individual bonds are polar?

• State the role of a catalyst in a reaction pathway and whether it changes the equilibrium constant.

Answer Key

• 1-B, 2-A, 3-B, 4-C, 5-B.

• Numerical checks: 0.25 M; 2.5 L; reduce subscripts to the simplest whole-number ratio; molarity becomes half; oxidation occurs at the electron-releasing electrode.

- Organic answers should mention π bonding/addition, delocalization, reaction type, vector cancellation/geometry, and lower activation energy without changing equilibrium.